॥वसुधैव कुटुम्बकम्॥

# SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

**Symbiosis International (Deemed University)**

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**Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)**

## Assignment No. 11

| | |
|---|---|
| **Subject:** | DevOps Lab |
| **TCODE:** | TE7950 |
| **Name of Student** | **Yash Patel** |
| **PRN No.** | **23070122235** |
| **Branch** | CSE, Batch (2023-27), L3 |
| **Academic Year & Semester** | 25 – 26, Sem VII |
| **Date of Performance** | 23rd July, 2026 |
| **Title of Assignment:** | Docker Compose for Multi-Container App |

**Theory:**

Docker Compose is a tool used to define and manage multi-container Docker applications using a single docker-compose.yml file. It allows multiple services, such as a web application and a database, to be configured, started, and managed together with a single command. Docker Compose automatically creates a network for communication between services, manages named volumes for persistent storage, and simplifies deployment by eliminating the need to start each container individually. This makes it easier to develop, test, and maintain applications that consist of multiple interconnected containers.

**Commands Used**

- Created a new project directory and navigated into it:
  ```
  mkdir my-multi-app
  cd my-multi-app
  ```
- Started the multi-container application in detached mode:
  ```
  docker compose up -d
  ```
- Verified the running containers:
  ```
  docker compose ps
  ```
  *(or docker ps if docker compose ps was not available)*
- Stopped and removed the containers and network created by Docker Compose:
  ```
  docker compose down
  ```

**Step 1: Create the Project Directory**

Created Project Directory — Created a new project directory named my-multi-app and navigated into it to store the Docker Compose configuration.

**Step 2: Create docker-compose.yml**

Created Docker Compose Configuration — Created and saved the docker-compose.yml file defining the Flask web service, Redis service, Docker volumes, and networking configuration for the multi-container application.

**Step 3: Start the Multi-Container Application**

Started Multi-Container Application — Started the multi-container application using Docker Compose, which created the required network, volumes, and launched both the Flask web service and Redis service.

**Step 4: Access the Web Application**

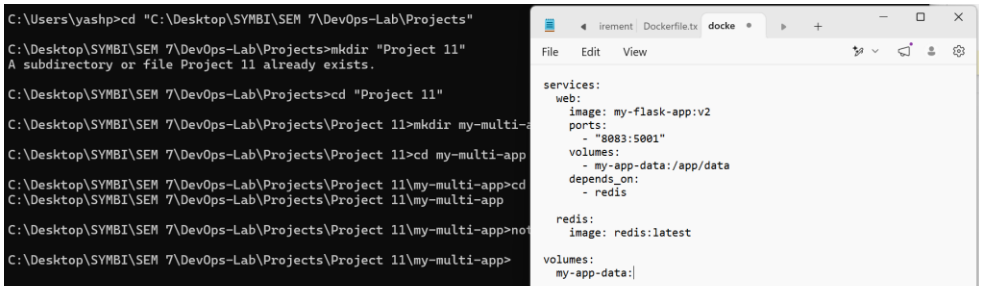
Verified Web Application — Accessed the Flask application through http://localhost:8083 and verified that the web service was successfully running as part of the Docker Compose application.

**Step 5: Verify the Running Services**

Verified Running Services — Verified that the Docker Compose application was running successfully by listing the active containers, including the Flask web service and the Redis service.

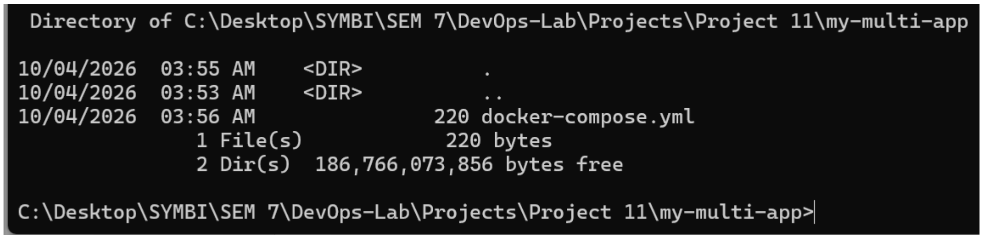
**Step 6: Stop and Remove the Multi-Container Application**

Stopped Multi-Container Application — Stopped and removed the Docker Compose application, including the running containers and the automatically created network, using the docker compose down command.

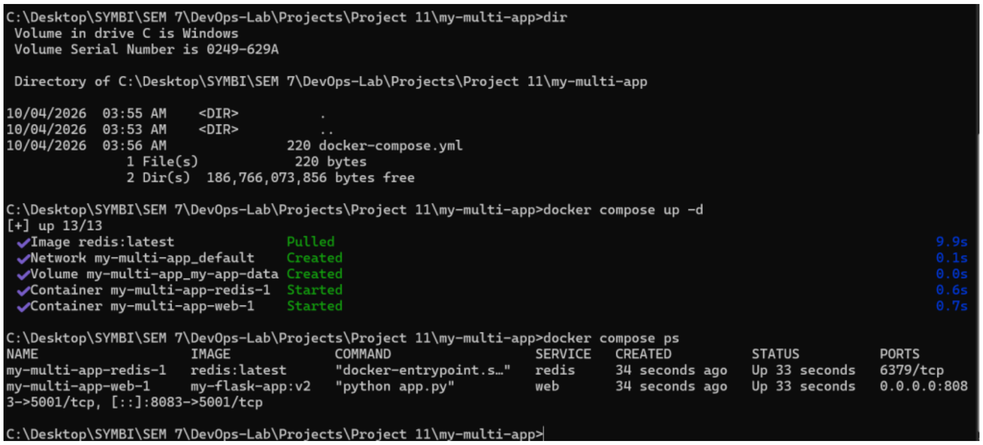


```
C:\Users\yashp>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>mkdir "Project 11"
A subdirectory or file Project 11 already exists.

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects>cd "Project 11"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11>mkdir my-multi-a

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11>cd my-multi-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>cd
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>not

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>
```

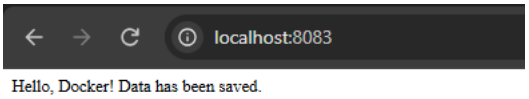
```
services:
  web:
    image: my-flask-app:v2
    ports:
      - "8083:5001"
    volumes:
      - my-app-data:/app/data
    depends_on:
      - redis

  redis:
    image: redis:latest

volumes:
  my-app-data:
```



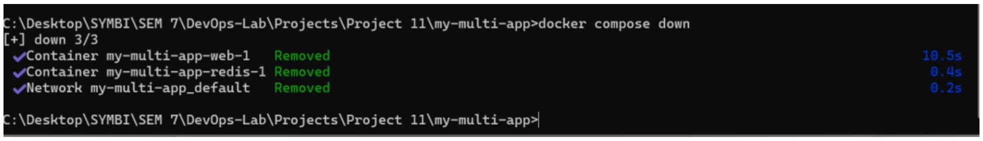
```
 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app

10/04/2026  03:55 AM    <DIR>          .
10/04/2026  03:53 AM    <DIR>          ..
10/04/2026  03:56 AM               220 docker-compose.yml
               1 File(s)            220 bytes
               2 Dir(s)  186,766,073,856 bytes free

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>
```



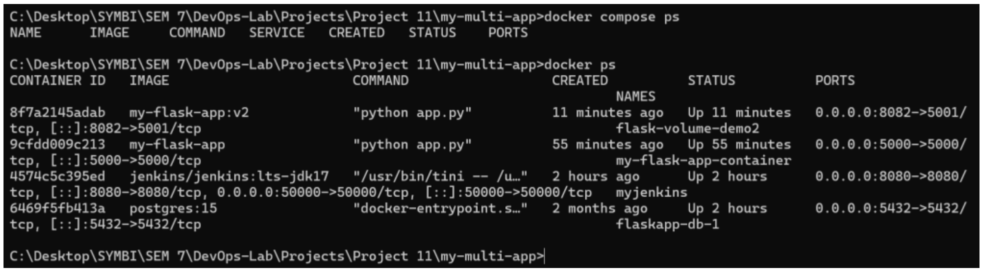
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 0249-629A

 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app

10/04/2026  03:55 AM    <DIR>          .
10/04/2026  03:53 AM    <DIR>          ..
10/04/2026  03:56 AM               220 docker-compose.yml
               1 File(s)            220 bytes
               2 Dir(s)  186,766,073,856 bytes free

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>docker compose up -d
[+] up 13/13
 ✓Image redis:latest             Pulled                                                          9.9s
 ✓Network my-multi-app_default    Created                                                         0.1s
 ✓Volume my-multi-app_my-app-data Created                                                         0.0s
 ✓Container my-multi-app-redis-1  Started                                                         0.6s
 ✓Container my-multi-app-web-1    Started                                                         0.7s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>docker compose ps
NAME                   IMAGE              COMMAND                    SERVICE   CREATED          STATUS          PORTS
my-multi-app-redis-1   redis:latest       "docker-entrypoint.s…"   redis     34 seconds ago   Up 33 seconds   6379/tcp
my-multi-app-web-1     my-flask-app:v2    "python app.py"            web       34 seconds ago   Up 33 seconds   0.0.0.0:808
3->5001/tcp, [::]:8083->5001/tcp

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>
```



Hello, Docker! Data has been saved.



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>docker compose down
[+] down 3/3
 ✓Container my-multi-app-web-1   Removed                                                          10.5s
 ✓Container my-multi-app-redis-1 Removed                                                           0.4s
 ✓Network my-multi-app_default   Removed                                                           0.2s

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>
```



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED          STATUS          PORTS
                                                                            NAMES
8f7a2145adab   my-flask-app:v2             "python app.py"          11 minutes ago   Up 11 minutes   0.0.0.0:8082->5001/
tcp, [::]:8082->5001/tcp                                                    flask-volume-demo2
9cfdd009c213   my-flask-app                "python app.py"          55 minutes ago   Up 55 minutes   0.0.0.0:5000->5000/
tcp, [::]:5000->5000/tcp                                                    my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17   "/usr/bin/tini -- /u…"   2 hours ago      Up 2 hours      0.0.0.0:8080->8080/
tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
6469f5fb413a   postgres:15                 "docker-entrypoint.s…"   2 months ago     Up 2 hours      0.0.0.0:5432->5432/
tcp, [::]:5432->5432/tcp                                                    flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 11\my-multi-app>
```

Subject Faculty Name: Prof. Arijit Dutta